



SMART INDIA HACKATHON 2026



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HACKATHON
2026

TITLE PAGE

- Problem Statement ID – **SIH26106**
- Problem Statement Title- **AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID-
- Team Name - **HEADWATERS**



TRACEVAULT

(Email Threat Detection & Forensic Intelligence)



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ARCHITECTURE

WHO USES



1. Gmail Users
Employees at banks,
govt. offices,
universities



2. IT / Security Admins
Manage security
policies and review
flagged summaries



3. SOC / Security Analysts
Investigate threats
and analyze
suspicious emails



4. Fraud / Incident Response Teams

HOW THEY USE



Use TraceMail Gmail Extension
Real-time scanning of
incoming emails



Access Admin Dashboard
Review alerts, set
policies, manage users
and rules



Open Alerts in TraceVault
Deep-dive into threat
details and context



Investigate & Export Report

PROBLEM EXISTING

- **Attackers use spoofed domains, deceptive sender identities, and AI-generated phishing language to bypass spam filters and blacklists.**
- **Existing tools don't correlate headers, SPF/DKIM/DMARC, IP reputation, geolocation, and domain data into one picture.**
- **Even after detection, organizations can't trace the source path or identify probable sender infrastructure.**



PROPOSED SOLUTION

- **AI-powered detection** using NLP, ML, and header forensics to flag phishing, spoofing, and impersonation in real time.
- **Origin traceability** through relay-path reconstruction, IP geolocation, and domain intelligence.
- **Graph-based correlation** linking related fraud emails into campaign-level attribution.
- **Forensic reporting** with confidence-based risk scoring for investigation and legal use.



UVP (UNIQUE VALUE PROPOSITION)

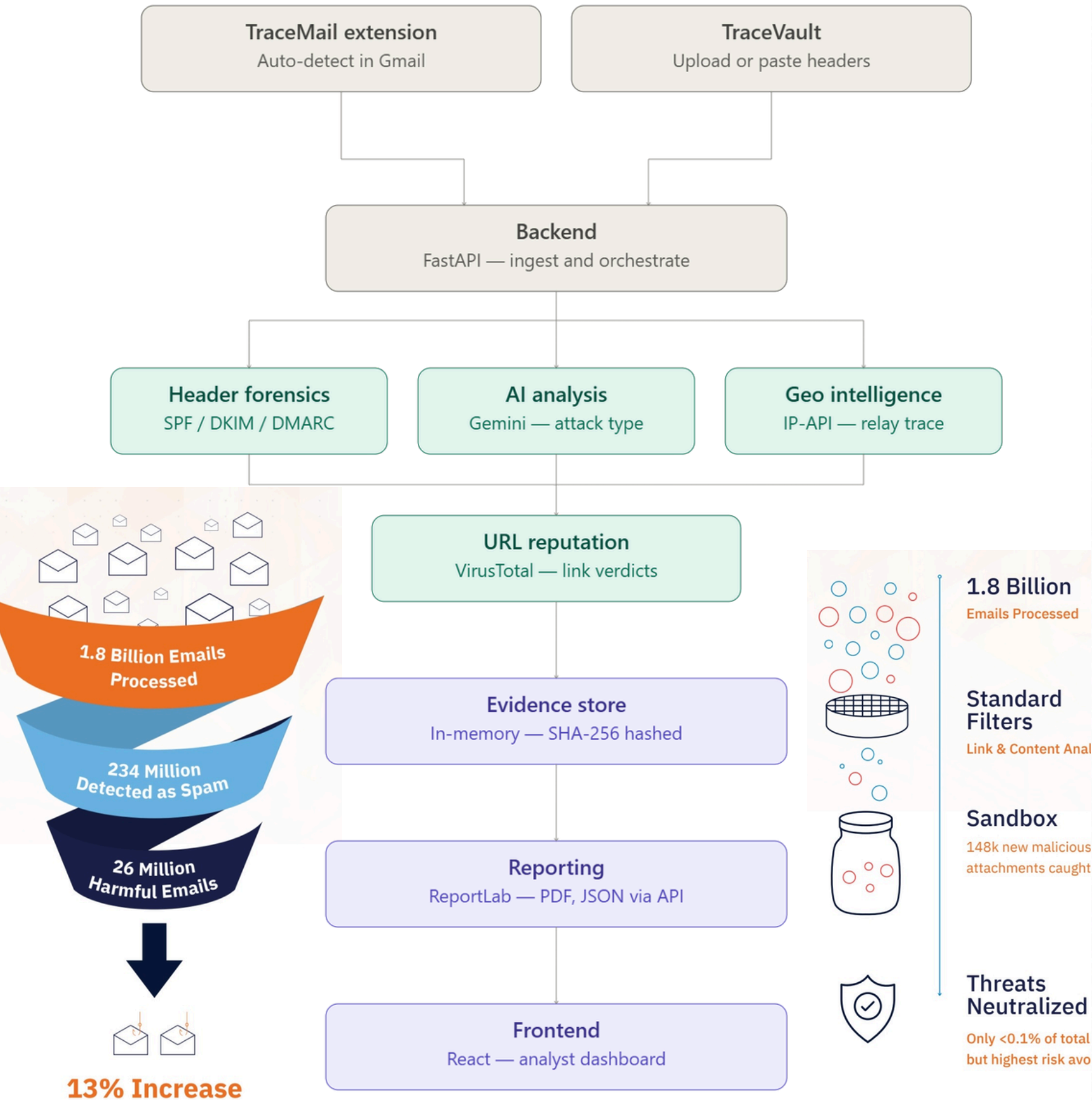
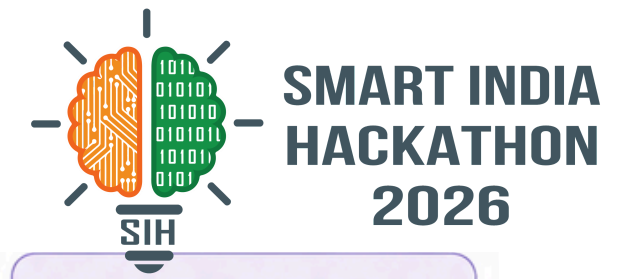
- **Detection where the user reads mail** — TraceMail lives inside Gmail, not a separate app users forget to check.
- **Investigation, not just a verdict** — TraceVault gives evidence and a "why," where legacy filters only give block/allow.
- **Confidence-scored geolocation** — traces infrastructure origin with an honest confidence band, never a false claim of pinpointing a person.
- **Forensic-grade, tamper-evident reports** — hash-chained exports.

How it addresses the Problem

- **TraceMail** - It is the gmail extension that watches your inbox
- **TraceVault** - the website that turns flagged mail into a forensic report



TECHNICAL APPROACH



TECH STACK

TRACEVAULT

FRONTEND

- React
- JavaScript
- Tailwind CSS
- React Router
- Recharts
- React Flow

BACKEND

- Python
- FastAPI
- Uvicorn

AI & INTELLIGENCE

- LangChain
- Google Generative AI
- Embeddings
- ChromaDB

IP / URL / DOMAIN INTELLIGENCE

- IP / URL / Domain Intelligence

EMAIL FORENSICS

- Email Forensics

REPORTING

- ReportLab

TRACEMAIL

- JavaScript

- Chrome Extension Manifest V3

- Chrome Extension APIs

- HTML

- CSS

- Fetch API

DEPLOYMENT

- Vercel
- Render
- GitHub

WHAT GETS OPENED



Inline Threat Alert

Shows risk level directly in Gmail before opening



Admin Dashboard

Alerts overview, policy settings, user & rule management



Case / Alert View

Fraud score, email trace, indicators, correlation graph



Forensic Report

Hash-chained PDF/JSON report with full evidence & indicators

FEASIBILITY

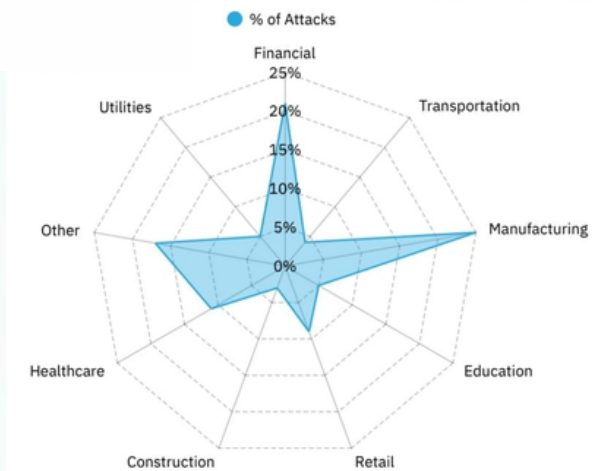
- Technical**
Python + NLP, email parsing, metadata extraction
- Prototype**
MVP flow → Analysis → Threat Score → Evidence → Report
- Operation**
Synthetic/authorized emails, safe datasets, mock validation

MVP Flow



VIABILITY

- Market**
 - Enterprises
 - Govt. organizations
 - Financial institutions
 - Educational institutions

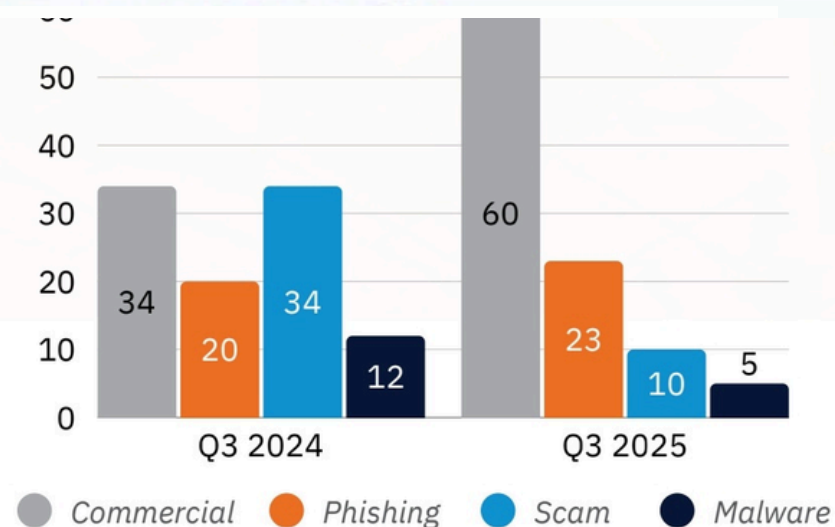


Investment & ROI

- Lower manual effort
- Faster triage
- Better accuracy
- Scalable & retained

TECHNICAL CHALLENGES

- Risk**
False positives, incomplete headers, IP limits, VPNs
- Mitigation**
Multi-signal scoring, confidence indicators, human review



BUSINESS / DEPLOYMENT CHALLENGES

- Risk**
Privacy & security controls during analysis, storage, investigation
- Mitigation**
Data minimization, RBAC, logs, retention, masking





OPERATIONAL IMPACT
 WHAT CHANGES FOR THE PERSON USING IT



Automated & Real-time
 Scans emails as they're opened — no manual checks.



Two-tier Triage
 Local heuristics first; AI + forensics only for real threats.



One Click to Evidence
 "Investigate in TraceVault" opens the full forensic report, no re-entry.



Bulk Investigation
 Re-investigate multiple flagged emails in one action.





WHO BENEFITS?
 REAL IMPACT FOR THE RIGHT TEAMS



Financial Institutions
 Phishing impersonating banks is your own demo's exact scenario.



Government & PSU email systems
 Matches AICTE's own theme framing (Cyber Security Cell).



SOC / incident response teams
 Bulk-investigate and case-history features exist specifically for this workflow.

KEY FEATURES & IMPACT		Existing solutions	TraceVault
	AI-powered threat classification	✗	✓
	SPF / DKIM / DMARC forensic breakdown	⚠	✓
	IP geolocation with confidence scoring	✗	✓
	VirusTotal URL reputation check	⚠	✓
	Automatic in-inbox detection	⚠	✓
	One-click investigation handoff	✗	✓
	Downloadable forensic PDF report	✗	✓
	Legal-ready evidence certificate (BSA 2023)	✗	✓
	Campaign correlation graph	✗	⚠



TECHNICAL FOUNDATION

- **Phishing Detection:** AI-based detection and stronger email authentication are needed to combat evolving email fraud. [nih-gov-articles](#) , [research paper](#)
- **Email Forensics:** Header, metadata, and email-tracing techniques are essential for investigations. [LinkedIn post](#)
- **Threat Prevention:** Layered detection against phishing, spoofing, spam, and targeted attacks improves email security. [IEEE](#)
- **Attack Classification:** Mapping phishing and Business Email Compromise techniques to the MITRE ATT&CK framework improves systematic threat categorization over manual analyst review. [BEC TTP review](#)
- **Network Forensics:** Combining IP geolocation, WHOIS domain lookup, and threat-intelligence blacklist checks strengthens attribution of malicious sending infrastructure. [arxiv SPECTRE](#)
- **Evidence Admissibility:** Indian courts have affirmed that hash-verified electronic records satisfy authentication requirements under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023. [chain of custody paper](#)

PROJECT RESOURCES

- **github repo link:** <https://github.com/bhavya1101-cse/tracevault>
- **vercel-frontend(TraceVault):** <https://tracevault-seven.vercel.app/>
- **chrome extension(TraceMail):** <chrome://extensions/>

